

CS221-2021-Part-B-30 Minutes

Quiz 4 CS221

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Points: 27/35

1

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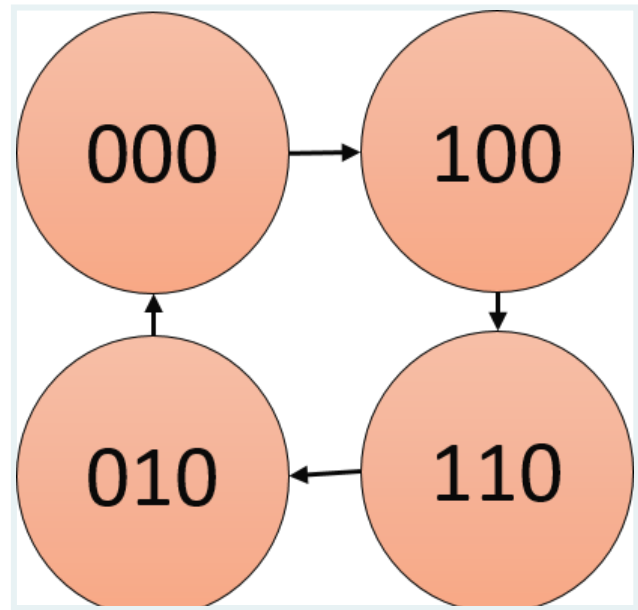
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✗ **Incorrect** 0/2 Points

4

=> Suppose a counter which count 0, 4, 6, 2 and repeat as shown in the diagram. This clearly require 2 bit state register to implement this and assume state S0, S4, S6 and S2 represent for the count 0, 4, 6 and 2 respectively.

Our main motivation is to minimize the Z logic but not the NS logic. Which of the encodings are better.



- ☒ S0(00), S2(01), S4(10), S6(11)
- ☒ S0(00), S2(10), S4(01), S6(11) ✓
- ☒ S0(01), S2(11), S4(00), S6(10) ✓
- ☐ S0(00), S2(11), S4(10), S6(01)

✓ **Correct** 1/1 Points

5

=> Pick the correct statement

- ☐ Characteristic table of a FF describe behavior of FF output for different inputs
- ☒ Excitation Table tabulate the required inputs for the desired output transitions ✓
- ☐ Both characteristic table and excitation table of a FF are same.
- ☐ None of the above

✓ **Correct** 2/2 Points

6

=> How many latches will be there in one Kilo Byte of memory with 10 bit address line and 8 bit data size (including the MAR and MBR).

16420

✓ **Correct** 1/1 Points

7

=> Which of the following assignment will convert D-flipflop to JK flipflop?

- ☒ $D = K'Q + JQ'$ ✓
- ☐ $D = KQ' + J'Q$
- ☐ $D = K'Q + J'Q$
- ☐ $D = KQ' + JQ'$

✓ **Correct** 2/2 Points

8

=> What is the minimum number of states required to design a Moore Machine which takes a 2-bit input (X, Y) and 2-bit output (Carry, Sum) such that output is the sum of the X, Y and Carry bits?

- ☐ 3
- ☒ 4 ✓
- ☐ 2

5

✓ **Correct** 1/1 Points

9

==> Choose the true option

- ☐ Some Mealy machines has an equivalent Moore machine
- ☒ Any Moore machine has an equivalent Mealy machine and vice versa ✓
- ☐ Some Moore machines has equivalent Mealy machine but not for all.
- ☐ All of the above

✗ **Incorrect** 0/1 Points

10

==> Is it possible to make any flip flop as universal flip? You are allowed to use any other circuitry/GATEs.

- ☐ Some flip flop can be made
- ☐ All flip flop can be made ✓
- ☒ Only JK FF can be made universal FF.
- ☐ non of the above

✓ **Correct** 2/2 Points

11

Benefit of Asynchronous Counter

- ☐ Easy to Design

- ☐ Can be used in slower circuit
- ☐ Can be used in smaller standalone circuit
- ☐ Smaller Circuitry
- ☒ All the above ✓

✓ **Correct** 2/2 Points

12

==> Suppose for a FSM, let the two states S1 and S2 of the FSM, if two transitions from these two states have a common next state S3 in the same FSM, then the best encoding for the states X, Y and Z among these following encodings will be:

- ☐ S1(11001), S2(11001), S3(00000)
- ☐ S1(10110), S2(11001), S3(11011)
- ☒ S1(10000), S2(00000), S3(11011) ✓
- ☐ S1(10000), S2(11111), S3(00000)

✓ **Correct** 3/3 Points

13

==> After minimization the given FSM of the Table, the number of states remain is :

PS	NS		Output
	X=0	X=1	
A	D	C	0
B	F	H	0
C	E	D	1
D	A	E	0
E	C	A	1
F	F	B	1
G	B	H	0
H	C	G	1

6

✗ **Incorrect**

14

=> How many flip-flops will be required in state register to implement a counter which count the sequence 6, 12, 8, 12, 18, 13, 112, 127 and repeat the same.

4

Correct answers:

3

✗ **Incorrect** 0/1 Points

15

Race condition occurs?

- ☐ Both Q and Q' set to 0 of a RS Latch ✓
- ☐ Q set to 1 and Q' set to 0
- ☐ When J=1 and K=1 in JK Flip-flop.
- ☒ non of the above

✓ **Correct** 2/2 Points

16

=> Given a FSM with 32 states. Implementation of the FSM requires x, y and z flip flops using binary encoding, one hot encoding and gray code encoding schemes respectively, then the value of $x + y + z$ will be _____.
(Write one positive integer answer)

42

✓ **Correct** 3/3 Points

17

==> For the given FSM in the state table, determine the equivalence states using any methods

<i>PS</i>	<i>NS</i>		<i>Z</i>	
	<i>X=0</i>	<i>X=1</i>	<i>X=0</i>	<i>X=1</i>
<i>A</i>	<i>B</i>	<i>C</i>	<i>0</i>	<i>1</i>
<i>B</i>	<i>A</i>	<i>E</i>	<i>1</i>	<i>0</i>
<i>C</i>	<i>D</i>	<i>A</i>	<i>0</i>	<i>1</i>
<i>D</i>	<i>C</i>	<i>E</i>	<i>1</i>	<i>0</i>
<i>E</i>	<i>A</i>	<i>F</i>	<i>1</i>	<i>0</i>
<i>F</i>	<i>E</i>	<i>F</i>	<i>0</i>	<i>1</i>

- ☐ (A, B) and (C, D)
- ☒ (A, C) and (B, D) ✓
- ☐ (A, D) and (B, C)
- ☐ (A,C,D) and B

✓ **Correct** 1/1 Points

18

Why using JK flip flpp in State Register of FSM will be beneficial as compared to D- Flip flop?

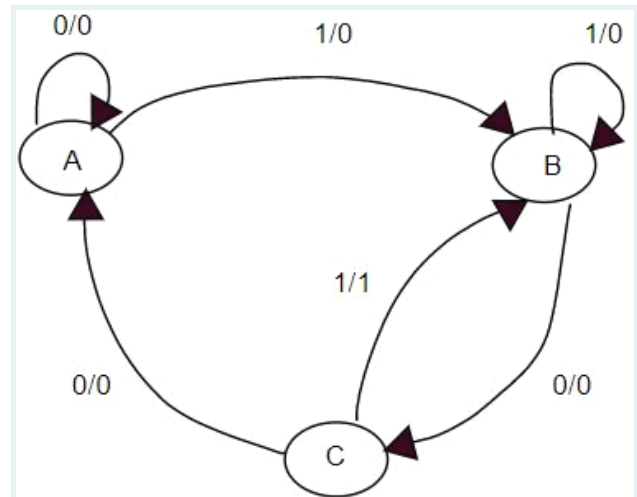
- ☐ JK Flip have two inputs
- ☐ D-FF has only one input, which makes bigger circuit

- ☒ Excitation table JK flip flop have around 50% do not care ✓
- ☐ Non of the above

✗ **Incorrect** 0/2 Points

19

==>A FSM with one input X and one output Z is defined by the state diagram below. Assume the FSM starts in state A. State register output as S1 and S0 (encoded serially). What will be Z logic and NS logic to set D1 and D0?



- ☐ $Z=S1'.S0'.X$ $D1=S1.S0'.X'$ $D0=S1+S0$
- ☐ $Z=S1.S0'.X$ $D1=S1'S0.X'$ $D0=(S1.S0)'$
- ☐ $Z=S1.S0'.X$ $D1=S1'.S0.X'$ $D0=(S1.S0)'X$ ✓
- ☒ None of the above

✓ **Correct** 3/3 Points

20

==>For the given FSM in the state table, find which states are equivalent.

PS	NS X=0,1	Z/OP 0,1
S0	S1, S3	0,1
S1	S2, S2	0,1
S2	S0, S3	1,0
S3	S0, S2	1,0

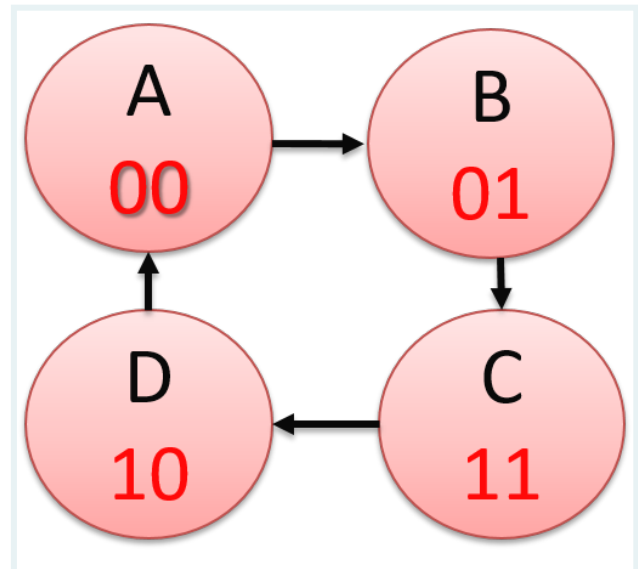
- ☐ (S0, S1)
- ☒ (S2, S3) ✓

- ☐ (S0, S3)
- ☐ None of the above

✗ **Incorrect** 0/1 Points

21

=>How many extra gates are required to implement given FSM (assume state FF of state register gives complemented output).



- ☐ 0 ✓
- ☒ 1
- ☐ 2
- ☐ 3

✗ **Incorrect** 0/1 Points

22

=>We want to design a 4 bit multi-function register with these five functions: (a) Right Shift with rotate, (b) Left shift with rotate, (c) retain the older value, (d) Complement the value, and (f) parallel out. What will be required size of multiplexors for every bit of the register to design this multi function register.

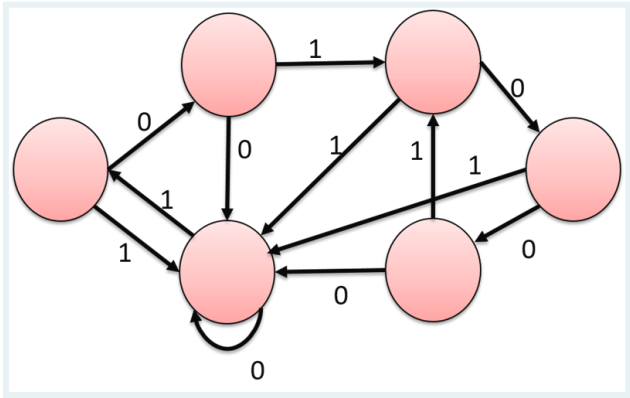
- ☐ 8 to 1 Mux
- ☒ 4 to 1 Mux ✓

- ☐ 2 to 1 Mux
- ☒ Non of the above

✓ Correct 2/2 Points

23

==>Choose the correct option regarding the following FSM



- ☒ It is correct and complete ✓
- ☐ It is not correct but complete
- ☐ It is correct but not complete
- ☐ It is not correct and not complete

✓ Correct 2/2 Points

24

==>For the given FSM in the state table, find which states are equivalent

PS	NS X=0,1	Z/OP 0,1
S0	S1, S2	1,0
S1	S3, S1	0,0
S2	S0, S3	1,0
S3	S2, S1	1,0

- ☐ S0, S1
- ☐ S2, S3
- ☐ S0, S3

☐ non of the above ✓

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